

1. SPECIFICATION METADATA

Use Case ID: UC-03	Use Case Name: Redeem Digital Gift Card with Anti-Fraud Voucher Locking
Primary Actor: Account Holder	Secondary Actors: Merchant Partner API, Anti-Fraud & Risk Engine
Description: Account Holder redeems universal wallet credits for a merchant digital gift card, backed by real-time risk scoring, anti-fraud voucher locking, and merchant ledger reconciliation.	

2. PRECONDITIONS & POSTCONDITIONS

Preconditions: - Account Holder is authenticated with an active session. - Account Holder possesses sufficient universal credit balance. - Target Merchant Partner API is active and reachable via mTLS.	Postconditions: - Universal credits permanently debited from user wallet ledger. - Single-use gift card voucher issued with SHA-256 checksum. - Merchant Partner settlement ledger balance credited. - Tamper-evident transaction logged to immutable audit ledger.
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3. MAIN SUCCESS SCENARIO (STEP-BY-STEP)

- Account Holder** browses the digital wallet catalog and selects a Merchant Partner and gift card denomination (\$50).
- System** calculates required universal exchange credits, displays conversion quote, merchant terms, and expiration.
- Account Holder** confirms the gift card redemption order.
- System** verifies that available universal exchange balance is equal to or exceeds the redemption cost.
- System** transmits transaction telemetry (User ID, Device ID, IP, Velocity, Amount) to the Anti-Fraud Engine.
- Anti-Fraud & Risk Engine** analyzes telemetry, calculates a low risk score (<30), and approves immediate issuance.
- System** places a temporary transactional hold on the required universal exchange credits in the user wallet.
- System** sends an authenticated issuance payload to the Merchant Partner API via secure TLS 1.3.
- Merchant Partner API** verifies inventory and returns an authorized raw digital voucher payload.
- System** generates SHA-256 cryptographic checksum, locks voucher to Account Holder ID, permanently debits credits, credits merchant ledger, and renders the scannable QR/barcode on screen.

4. ALTERNATE FLOWS & EXCEPTION HANDLING

Alternate Flow 4a: Insufficient Universal Credits 4a1. At Step 4, system detects available balance is less than required denomination. 4a2. System alerts user of deficit and offers to trigger UC-02: Convert Points to Universal Credits from linked merchant programs. 4a3. Use case terminates safely with zero ledger modification.	Alternate Flow 6a: High Fraud Risk Detected — Anti-Fraud Voucher Lock Triggered (Stereotype: «extend») 6a1. At Step 6, Anti-Fraud Engine detects anomalous pattern (unfamiliar IP/device, high velocity) and returns Risk Score ≥ 70. 6a2. System triggers UC-06: Apply Anti-Fraud Voucher Lock, locking voucher status to 'Locked-Pending Verification'. 6a3. System prompts Account Holder for Step-Up Biometric / OTP Multi-Factor Authentication. 6a4. If MFA succeeds within 60s, voucher lock clears and transaction resumes at Step 7 of Main Success Scenario. 6a5. [Flow 6b - MFA Failure]: If MFA fails or times out, system aborts transaction, releases credit holds, and flags incident.
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5. QUALITY & PERFORMANCE CONSTRAINTS

Latency: Entire pipeline (Steps 1–10) terminates in ≤ 250 ms (NFR-001) with zero ledger drift. | **Security:** All voucher payloads encrypted via AES-256 and signed via HMAC-SHA256 (NFR-002).